

RESEARCH EXPERIENCE

CERN Summer Studentship

30/6/2025 - 19/9/2025

- Contributions made to the open source simulation toolkit Garfield++ in order to model the space charge effect in microscopic tracking simulations. This work enables more accurate simulations of gaseous and silicon detectors, and improved understanding of the physical processes underlying detector behaviour.
- Framework created to model the space charge field semi-analytically in avalanche simulations. This framework was applied to investigate whether resistive anodes in gaseous detectors significantly affect the space charge field. Investigations were made into the charge densities required to significantly affect detector behaviour. Results were applied in a simulation study, confirming the presence of a gain-limiting effect. Model applied to simulate the space charge effect in Micromegas detectors.
- Progress regularly presented to the CERN GDD group. Seminar presentation given to the CERN EP-DT group. Project report written, and added as a DRD1 technical note. Work presented at the Oct. 2025 DRD1 collaboration meeting (Warsaw, Poland).
- Supervisors: Djunes Janssens, Heinrich Schindler.

Skills:

- Languages: C++, Python, Bash. Tools: Garfield++, Git, HTCondor, lldb, Valgrind/Callgrind. General: techniques for code optimisation and efficiency, object-oriented programming, operational principles of gaseous detectors, presentation skills, report writing.

STFC Summer Studentship

1/7/2024 - 23/8/2024

- Investigation made into the use of Garfield++ and Magboltz for tracking the production of scintillation light in CF₄/Ar mixtures. Application to the GEM geometry and E-field configuration present in the MIGDAL detector. A proof of concept scintillation tracking system was developed and results compared to experiment data. Methods for reducing computation time of GEM drift simulations were investigated.
- Progress regularly presented to the collaboration. Seminar presentation given to researchers and other summer students. Progress regularly shared with the CERN GDD group. Work presented at a CERN DRD1 WG4 Working Meeting. Project report assembled.

Skills:

- Languages: C++, Python, Bash. Tools: Git, HTCondor, Gmsh, Elmer FEM, Garfield++, Magboltz. General: object-oriented programming, operational principles of GEMs and TPCs, presentation skills, report writing.

PROJECTS

MPhys Project

10/2025 - 4/2026

- Investigated the particle ID (PID) performance of the $K_s \rightarrow \pi\pi$ calibration sample for the LHCb experiment. Identified and removed contributions from background processes. Fits produced with RooFit. The sPlot technique was applied to unfold signal from background. PID efficiencies and misidentification rates were computed; comparison made between two commonly used measures of PID confidence (DLL and ProbNN). Results compared to those of the $D^{*0} \rightarrow D^0\pi^+$ calibration sample. Project report assembled.

Skills:

- Languages: Python, Bash. Tools: ROOT, RooFit, HTCondor. General: operational principles of the LHCb experiment, report writing.

- Investigation into the properties of open stellar clusters. Gained understanding of techniques for processing astronomical images, including data reduction and photometry. Membership to clusters was confirmed by matching proper motion and parallax data from the Gaia satellite to the processed data using TOPCAT. Colour magnitude diagrams were produced, and isochrones fitted to them. Age, distance and extinction parameters, with estimated errors, were obtained. Literature search carried out to test agreement of results. Results presented in a scientific report, which scored 85.

Skills:

- Tools: QFitsView, SAOImageDS9, TOPCAT. General: rapidly learning new scientific software, report writing.

 N -Body Simulation

2024

- Numerical simulation of N bodies interacting under gravity. Velocity Verlet algorithm used. Supports an arbitrary number of bodies with user-defined masses and initial positions/velocities. Capability to track energy and momentum conservation.

Skills:

- Principles of many-particle simulations. Numerical methods for simulating dynamical systems. Writing efficient, vector based code in Python.

EDUCATION

University of Oxford, Oriel College (MPhys Physics)

2021-2026

- First Class degree classification in Bachelor's component of MPhys. Academic Scholarship (prize for academic performance) awarded in both second and third year of studies. Prizes awarded for performance in all internal college exams to date.

Courses: Quantum mechanics, thermodynamics and statistical mechanics, electromagnetism, mathematical methods (second year); classical mechanics, fluids, condensed matter physics, nuclear and particle physics, symmetry and relativity, general relativity (third year); theoretical physics (including QFT), particle physics (fourth year).

PERSONAL

Teaching	129 pages of $\text{\LaTeX}$ -formatted notes on quantum mechanics and linear algebra were created and distributed to students in years below. Experienced in tutoring (GCSE & A Level maths).
Languages	English (fluent), Spanish ($\sim$ C1).
Music	Guitar, bass, drums. Production, songwriting and performance.
Sport	Running, squash, gym.